



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

09-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,B,C	A,C,D	B,C	A,C	A,B	A,C,D	C,D	A,B	A,B
SECTION-II (i)	Q.	1	2	3	4	5					
	A.	1.00	5.00	4.00	2.00	2.00					
SECTION-II (ii)	Q.	6	7	8	9	10					
	A.	4	2	4	2	6					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,B,C,D	B,C	A,B,C	A,B,C,D	C,D	A,B,C	B,C	A,D	A,C,D
SECTION-II (i)	Q.	1	2	3	4	5					
	A.	4.00	137.76	-241.96 to -241.97	0.20	5.00					
SECTION-II (ii)	Q.	6	7	8	9	10					
	A.	3	6	6	2	3					

PART-3 : MATHEMATICS

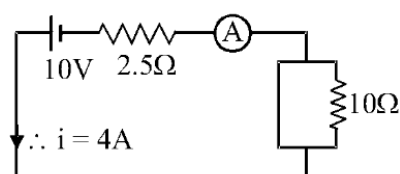
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,D	A,B,C,D	A,B,D	A,B	A,D	A,D	A,C,D	A,B,D	B,C,D	A,B
SECTION-II (i)	Q.	1	2	3	4	5					
	A.	33.33	1.50	3.33	20.00	48.78					
SECTION-II (ii)	Q.	6	7	8	9	10					
	A.	1	1	3	180	4					

HINT – SHEET

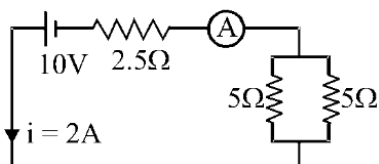
PART-1 : PHYSICS

SECTION-I

1. Ans (B,C,D)



Maximum



Minimum

2. Ans (A,B,C)

The current is maximum when the charge on the capacitor is zero

Current and charge are exactly 90 degrees out of phase in an ideal LC circuit (no resistance), so when the current is maximum the charge must be identically zero.

3. Ans (A,C,D)

Along X axis, $u_x = u \cos \theta$,

Along Y axis, $u_y = u \sin \theta$, $a_y = -g$

Equations of motion along X and Y axes would be

$$x = u \cos \theta t - \frac{qE}{2m} t^2 \text{ and } y = u \sin \theta t - \frac{1}{2} g t^2$$

Time of flight remains unchanged as vertical motion is not affected by $\vec{E}$. Range of the particle

in the present case is always less than $\frac{u^2 \sin 2\theta}{g}$

whatever be the value of E.

$$R = x_{T=T} = u \cos \theta \times \frac{2u \sin \theta}{g} - \frac{1}{2} \times \frac{qE}{m} \left[\frac{2u \sin \theta}{g} \right]^2 < \frac{u^2 \sin 2\theta}{g}$$

4. Ans (B,C)

$$\frac{F}{2\pi x \ell} = \eta \frac{dv}{dx}$$

$$\frac{F}{2\pi \ell} \int_R^{2R} \frac{dx}{x} = \eta \int_{-v_0}^{2v_0} dv$$

$$\frac{F}{2\pi \ell} \times \ln 2 = \eta \times 3v_0$$

$$\frac{F}{\ell} = \frac{6\pi v_0 \eta}{\ln 2}$$

$$\frac{F}{2\pi \ell} \times \ln \left(\frac{x}{R} \right) = \eta v_0$$

$$\Rightarrow \frac{3}{\ln 2} \ln \left(\frac{x}{R} \right) = 1$$

$$\Rightarrow \ln \left(\frac{x}{R} \right) = \ln 2^{1/3} \Rightarrow x = 2^{1/3} R$$

5. Ans (A,C)

$$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_\ell}; f_m = -\frac{15}{2} \text{ cm}$$

$$\frac{1}{f_\ell} = \frac{1}{f_g} + \frac{1}{f_w}$$

$$= \left(\frac{3}{2} - 1 \right) \left(\frac{1}{-10} - \frac{1}{15} \right) + \left(\frac{4}{3} - 1 \right) \left(\frac{1}{15} + \frac{1}{15} \right)$$

$$= \frac{-1}{12} + \frac{2}{45} = \frac{-7}{180}$$

6. Ans (A,B)

$$mg \sin \theta - T = ma \quad \dots (1)$$

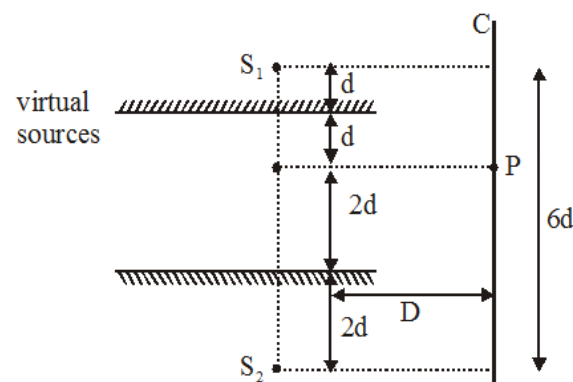
$$Tr = I\alpha \quad \dots (2)$$

$$r\alpha = a \quad \dots (3)$$

7. Ans (A,C,D)

Equilibrium position will be midway between -2 & 6 m i.e. 2 m and amplitude $= 6 - 2 = 4$ m

8. Ans (C,D)



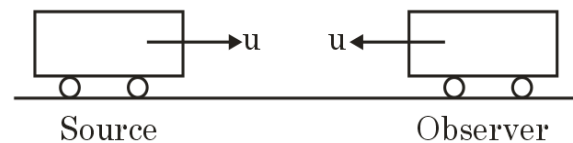
for P, $y = d$

$$\Delta x = d \cdot \frac{6d}{D} = \frac{6d^2}{D}$$

for constructive interference

$$\frac{6d^2}{D} = n\lambda$$

9. Ans (A,B)



Velocity of sound (v') $= v \pm w$ (depends on direction of flow of wind)

$$\text{Apparent frequency } f_2 = \left(\frac{v' + u}{v' - u} \right) f_1$$

$\Rightarrow f_2 > f_1$ in both cases

10. Ans (A,B)

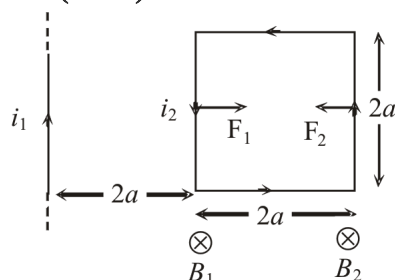
$$\begin{aligned} W_{AB} &= nR(T_B - T_A) \\ &= 2R(600 - 300) \\ &= 600 R \end{aligned}$$

$$\begin{aligned} W_{BC} &= nRT_B \ln \frac{P_B}{P_C} \\ &= (2)R(600) \ln 2 \\ &= 1200R \ln 2 \\ &= 1200R \times 0.7 \\ &= 840R \end{aligned}$$

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (1.00)**



$$\begin{aligned} F_{\text{net}} &= F_1 - F_2 \\ &= i_2 B_1 (2a) - i_2 (B_2) (2a) \\ &= \left(\frac{\mu_0}{4\pi} \frac{2i_1}{2a} \right) i_2 (2a) - \left(\frac{\mu_0}{4\pi} \frac{2i_1}{4a} \right) i_2 (2a) \\ &= \frac{2\mu_0}{4\pi} - \frac{\mu_0}{4\pi} = \frac{\mu_0}{4\pi} \\ \therefore \beta &= 1 \end{aligned}$$

2. **Ans (5.00)**

$$47.2 = 13.6Z^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \Rightarrow Z = 5$$

3. **Ans (4.00)**

$$\begin{aligned} \text{Released energy} &= 2 \times 4 \times 7.5 - 2 \times 1.5 - 7 \times 5 \\ &= 22 \text{ MeV} \end{aligned}$$

4. **Ans (2.00)**

P.E. will be maximum second time when phase will change by $300^\circ \left(2\pi - \frac{\pi}{3} \right)$

$$t = \frac{5}{6}T = \frac{5}{6} \times \left(\frac{2\pi}{\omega} \right) = \frac{1}{60}$$

$$120 \times \frac{1}{60} = 2$$

5. **Ans (2.00)**

$$W = \Delta KE = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\text{Here } W = \vec{F} \cdot \vec{d} = (3\hat{i} + 4\hat{j}) \cdot (\hat{i} + \hat{j}) = 3 + 4 = 7$$

$$\Rightarrow 7 = \frac{1}{2} \times 2 \times v_2^2 - \frac{1}{2} \times 2 \times (\sqrt{3})^2$$

$$\Rightarrow v_2^2 = 10 \Rightarrow v_2 = \sqrt{10} \text{ m/s}$$

PART-1 : PHYSICS

SECTION-II (ii)

6. **Ans (4)**

$$g' = \frac{64}{100}g, \quad g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$$

$$\frac{64}{100}g = \frac{g}{\left(1 + \frac{h}{R}\right)^2}, \quad \left(1 + \frac{h}{R}\right)^2 = \frac{100}{64}$$

$$1 + \frac{h}{R} = \frac{5}{4}, \quad \frac{h}{R} = \frac{1}{4}, \quad h = R/4$$

7. **Ans (2)**

In steady state current through battery

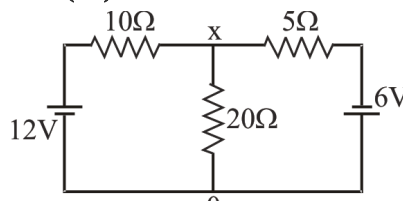
$$I = \frac{50}{23 + 3 + 17 + 7} = 1A$$

$$\Rightarrow Q_{10\mu F} = CV = 10 \times 20 = 200 \mu C$$

8. **Ans (4)**

$$v'_2 = (1 + e)v_{cm} - ev_2$$

9. **Ans (2)**



$$\frac{x}{20} + \frac{x+6}{5} + \frac{x-12}{10} = 0$$

$$\frac{2x + 8x + 48 + 4x - 48}{40} = 0$$

$$\Rightarrow x = 0 \Rightarrow i_1 = i_2$$

$$i_1 = \frac{12}{10} \Rightarrow i_1 + i_2 = \frac{12}{5}$$

10. **Ans (6)**

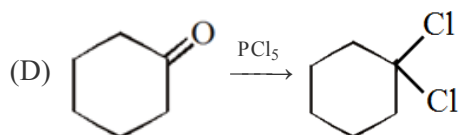
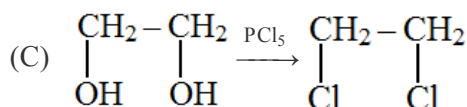
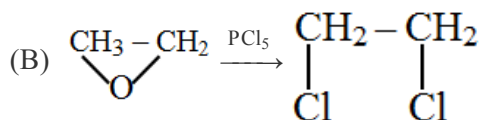
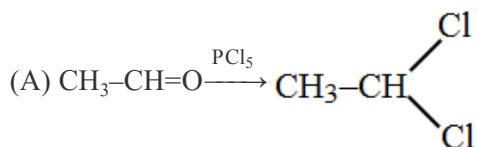
$$i = \sqrt{5}A$$

$$\frac{q_m^2}{2C} = \frac{q^2}{2C} + \frac{1}{2}Li^2 \Rightarrow q_{\text{max}} = 6C$$

PART-2 : CHEMISTRY

SECTION-I

3. Ans (B,C)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (4.00)

1, 3, 4, 5,

PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,D)

$$\mu^2 - 2\mu + 2 = 0$$

$$\therefore \alpha = 1 + i \quad \text{and} \quad \beta = 1 - i$$

$$x = \cot \theta - 1 \quad \therefore \quad x + \alpha = \cot \theta + i = \operatorname{cosec} \theta \cdot e^{i\theta}$$

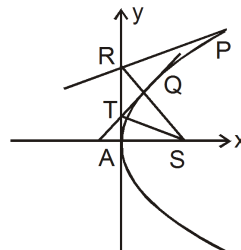
$$\text{and} \quad x + \beta = \cot \theta - i = \operatorname{cosec} \theta \cdot e^{-i\theta}$$

$$\alpha - \beta = 2i$$

$$\frac{(x + \alpha)^n - (x + \beta)^n}{\alpha - \beta} = \frac{\operatorname{cosec}^n \theta \cdot [e^{in\theta} - e^{-in\theta}]}{2i}$$

$$= \operatorname{cosec}^n \theta \cdot \frac{2i \sin n\theta}{2i} = \frac{\sin n\theta}{\sin^n \theta}$$

2. Ans (A,B,C,D)



$$R \equiv (1, 1)$$

$$T \equiv (2, 2)$$

$$\text{Equation of RS is } 4x - 3y - 1 = 0$$

$$\text{Equation of TS is } 3x - 2y - 2 = 0$$

$$\therefore \text{ focus } S \equiv (4, 5)$$

$$\text{length of latus rectum} = 4 \times \frac{1}{\sqrt{2}} = 2\sqrt{2}$$

$$\text{axis is } x + y - 9 = 0$$

$$\text{vertex} \equiv \left(\frac{9}{2}, \frac{9}{2} \right)$$

3. Ans (A,B,D)

$$\text{For the ellipse : } e = \sqrt{\frac{25-9}{25}} = \frac{4}{5}$$

$$\therefore \text{ foci are } (-4, 0) \text{ and } (4, 0)$$

For the hyperbola

$$ae = 4, e = 2$$

$$\therefore a = 2$$

$$b^2 = 4(4-1) = 12$$

$$b = \sqrt{12}$$

4. Ans (A,B)

$$I_n = \int_0^1 \frac{dx}{(1+x^2)^n} = \int_0^1 (1+x^2)^{-n} dx$$

$$= \frac{x}{(1+x^2)^n} \Big|_0^1 - \int_0^1 (-n)(1+x^2)^{-n-1} \cdot 2x \cdot x dx$$

$$= \frac{1}{2^n} + 2n \int_0^1 \frac{x^2}{(1+x^2)^{n+1}} = \frac{1}{2^n} + 2n \int_0^1 \frac{1+x^2-1}{(1+x^2)^{n+1}} dx$$

$$= \frac{1}{2^n} + 2n I_n - 2n I_{n+1}$$

$$\therefore 2n I_{n+1} = 2^{-n} + (2n-1) I_n$$

$$\therefore 2I_2 = \frac{1}{2} + I_1 = \frac{1}{2} + \tan^{-1} x \Big|_0^1$$

$$I_2 = \frac{1}{4} + \frac{\pi}{8}$$

5. **Ans (A,D)**

$$F'(x) = -\frac{2}{x^3} \int_4^x \{4t^2 - 2F'(t)\} dt + \frac{1}{x^2} (4x^2 - 2F'(x))$$

Put $x = 4$

$$F'(4) = 0 + \frac{1}{16} [4(16) - 2F'(4)] = 4 - \frac{F'(4)}{8}$$

$$\Rightarrow F'(4) = \frac{32}{9} \therefore \text{Option (A) is correct}$$

For option (D)

$$F(x) = \frac{1}{x^2} \int_4^x [4t^2 - 2F'(t)] dt$$

$$= \frac{1}{x^2} \left[\frac{4t^3}{3} - 2F(t) \right]_4^x$$

$$\therefore F(8) = \frac{1}{8^2} \left[\frac{4 \cdot 8^3}{3} - 2F(8) - \frac{4 \cdot 4^3}{3} + 2F(4) \right]$$

$$= \frac{32}{3} - \frac{F(8)}{32} - \frac{4}{3} + 2(0)$$

$$\therefore F(8) = \frac{32}{33} \left\{ \frac{28}{3} \right\}$$

$$\therefore F'(4) = \frac{11F(8)}{28}$$

6. **Ans (A,D)**

$$\frac{dy}{dx} (x^2 y^3 + xy) = 1 \Rightarrow \frac{dx}{dy} = x^2 y^3 + xy \Rightarrow \frac{dx}{dy}$$

$$-xy = x^2 y^3$$

$$\Rightarrow -\frac{1}{x^2} \frac{dx}{dy} + \frac{y}{x} = -y^3$$

put, $\frac{1}{x} = t$ and differentiating with respect to

$$y - \frac{1}{x^2} \frac{dx}{dy} + \frac{dt}{dy}$$

$$\Rightarrow \frac{dt}{dy} + ty = -y^3 \Rightarrow e^{\frac{y^2}{2}} t = \int -y^3 e^{y^2/2} dy$$

$$\Rightarrow \frac{1}{x} = 2 - y^2 + c e^{-\frac{y^2}{2}}$$

7. **Ans (A,C,D)**

$$f(x) = (x-1)4(x-2)^n, n \in \mathbb{N} \dots (1)$$

$$\therefore f'(x) = 4(x-1)^3(x-2)^n + (x-1)^4 n(x-2)^{n-1}$$

$$= (x-1)^3(x-2)^{n-1}(4x-8+nx-n)$$

$$= (x-1)^3(x-2)^{n-1}[(n+4)x - (n+8)]$$

If n is odd, then $f'(x) > 0$ if $x < 1$ and sufficiently close to 1 and $f'(x) < 0$ if $x > 1$ and sufficiently close to 1

$\therefore x = 1$ is point of local maximum

Similarly if n is even, then $x = 1$ is a point of local minimum

Further if n is even, then $f'(x) < 0$ for $x < 2$ and sufficiently close to 2 and $f'(x) > 0$ for $x > 2$ and sufficiently close to 2.

$\therefore x = 2$ is a point of local minimum.

8. Ans (A,B,D)

$$A^2 - 4A - 5I_3 = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$-4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} + \begin{bmatrix} -4 & -8 & -8 \\ -8 & -4 & -8 \\ -8 & -8 & -4 \end{bmatrix} + \begin{bmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

$$\therefore A^2 - 4A - 5I_3 = 0$$

$$\text{or } A^{-1} A^2 - 4A^{-1} A - 5A^{-1} I_3 = 0 \text{ or } (A^{-1} A) A - 4I_3 - 5A^{-1} = 0 \text{ or } IA - 4I_3 - 5A^{-1} = 0$$

$$\therefore A^{-1} = (A - 4I_3)$$

$$\text{Also, } |A^2| = \begin{vmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{vmatrix}$$

$$= 9(81 - 64) - 8(72 - 64) + 8(64 - 72)$$

$$= 9 \times 17 - 8 \times 8 + 8 \times (-8) = 133 - 128 = 5 \neq 0$$

$$\therefore A^2 \text{ is invertible}$$

$$\text{and } A^3 = A \cdot A^2 = A \cdot (4A - 5I_3) = 4A^2 - 5A$$

$$= \begin{bmatrix} 36 & 32 & 32 \\ 32 & 36 & 32 \\ 32 & 32 & 36 \end{bmatrix} + \begin{bmatrix} -5 & -10 & -10 \\ -10 & -5 & -10 \\ -10 & -10 & -5 \end{bmatrix} = \begin{bmatrix} 31 & 22 & 22 \\ 22 & 31 & 22 \\ 22 & 22 & 31 \end{bmatrix}$$

$$\therefore |A^3| \neq 0$$

$$\therefore A^3 \text{ is invertible.}$$

9. Ans (B,C,D)

$$x + 3y + 2z = 6 \quad \dots(i)$$

$$x + \lambda y + 2z = 7 \quad \dots(ii)$$

$$x + 3y + 2z = \mu \quad \dots(iii)$$

(A) If $\lambda = 2$, then $D = 0$, therefore unique solution is not possible

(B) If $\lambda = 4$, $\mu = 6$

$$x + 3y = 6 - 2z$$

$$x + 4y = 7 - 2z$$

$$\therefore y = 1 \text{ and } x = 3 - 2z$$

substituting in equation (iii)

$$3 - 2z + 3 + 2z = 6 \text{ is satisfied}$$

$\therefore$ infinite solutions

(C) $\lambda = 5$, $\mu = 7$

consider equation (ii) and (iii)

$$x + 5y = 7 - 2z$$

$$x + 3y = 7 - 2z$$

$$\therefore y = 0 \quad x = 7 - 2z \text{ are solution}$$

sub. in (i)

$$7 - 2z + 2z = 6 \text{ does not satisfy}$$

$\therefore$ no solution

(D) if $\lambda = 3$, $\mu = 5$

then equation (i) and (ii) have no solution

$\therefore$ no solution

10. Ans (A,B)

Equation of required plane is

$$\ell x + my + \lambda z = 0 \quad \dots(i)$$

angle between (i) & $\ell x + my = 0$ is α .

$$\Rightarrow \cos \alpha = \frac{\ell^2 + m^2}{\sqrt{\ell^2 + m^2} \sqrt{\ell^2 + m^2 + \lambda^2}}$$

$$\Rightarrow \cos^2 \alpha = \frac{\ell^2 + m^2}{\ell^2 + m^2 + \lambda^2} \Rightarrow \lambda = \pm \sqrt{\ell^2 + m^2} \tan \alpha$$

Hence equation of plane is

$$\ell x + my \pm z \sqrt{\ell^2 + m^2} \tan \alpha = 0$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (33.33)

$$f'(x) = \cos x - 2a \cos 2x - \cos 3x + 2a \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \cos x - \cos 3x + 2a(1 - \cos 2x) \geq 0$$

$$\Rightarrow 2 \sin 2x + \sin x + 4a \sin^2 x \geq 0$$

$$\Rightarrow 2 \sin^2 x (\cos x + a) \geq 0$$

$$a \geq -\cos x \Rightarrow a \geq 1$$

2. Ans (1.50)

Clearly $f'(x) = 0$

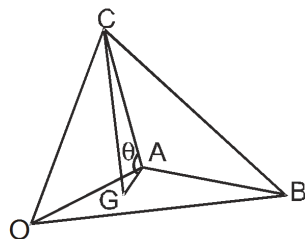
$$\therefore f(x) = c = 6$$

$$\therefore \sum_{r=1}^{25} f(r) = \sum_{r=1}^{25} 6 = 150$$

3. Ans (3.33)

Let $OABC$ be the tetrahedron.

Let G be the centroid of the face OAB ,



then $G A = \frac{1}{\sqrt{3}} AC$.

$$\text{Then } \cos \theta = \frac{GA}{CA} = \frac{1}{\sqrt{3}}$$

$$\therefore \cos^2 \theta = \frac{1}{3}$$

$$\therefore a = 1 \text{ and } b = 3$$

$$\therefore 10a + b = 13$$

4. Ans (20.00)

Total no. of arrangements = $15!$

Extreme chairs are occupied by girls, thus there are four gaps among 5 girls where boys can be seated.

Let the number of boys in these four gaps be $2x + 1$, $2y + 1$, $2z + 1$ and $2t + 1$, then

$$2x + 1 + 2y + 1 + 2z + 1 + 2t + 1 = 10$$

$$\Rightarrow x + y + z + t = 3$$

Where x, y, z, t are integers and $0 \leq x \leq 3, 0 \leq y \leq 3, 0 \leq z \leq 3, 0 \leq t \leq 3$

$\therefore$ The number of ways of selecting positions for boys

$$= \text{coefficient of } x^3 \text{ in } (1 + x + x^2 + x^3)^4 = \text{coefficient}$$

of x^3 in $\left(\frac{1-x^4}{1-x}\right)^4$

$$= \text{coefficient of } x^3 \text{ in } (1-x^4)^4 (1-x)^{-4} = {}^6C_3 = 20$$

$\therefore$ Number of arrangements of boys and girls with given condition $= 20 \times 10! \times 5!$

$$\therefore \text{Required probability} = \frac{20 \times 10! \times 5!}{15!} = \frac{20}{3003}$$

5. Ans (48.78)

There are $5W + 3R$ balls

Let E_1 be the event that first ball drawn is white
and E_2 be the event that first ball drawn is red

$$P(E_1) = \frac{5}{8} \quad P(E_2) = \frac{3}{8}$$

Let W be the event that second ball drawn is white

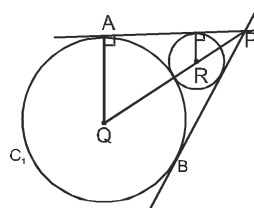
$$\therefore P(W/E_1) = \frac{4}{9} \text{ and } P(W/E_2) = \frac{7}{9}$$

$$\begin{aligned} P(E_1/W) &= \frac{P(E_1)P(W/E_1)}{P(W)} \\ &= \frac{P(E_1)P(W/E_1)}{P(E_1)P(W/E_1) + P(E_2)(W/E_2)} \\ &= \frac{5/8 \times 4/9}{5/8 \times 4/9 + 3/8 \times 7/9} = \frac{20}{20 + 21} = \frac{20}{41} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

6. **Ans (1)**



$$AQ = 3 + 2\sqrt{2}$$

$$PQ = 3\sqrt{2} + 4$$

Let 'r' be required radius

$$3\sqrt{2} + 4 = 3 + 2\sqrt{2} + r + r\sqrt{2}$$

$$\sqrt{2} + 1 = r(1 + \sqrt{2}) \Rightarrow r = 1$$

7. Ans (1)

Let line OPQ makes angle θ with x-axis so

$P \equiv (a \cos \theta, a \sin \theta)$, $Q (b \cos \theta, b \sin \theta)$

and Let $R(x, y)$

So $X = a \cos \theta$ $Y = b \sin \theta$

eliminating θ , we

get

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

locus of R is an

ellipse.

Also $a < b$ so

vertices are $(0, b)$

and $(0, -b)$

and extremities of minor axis are $(\pm a, 0)$.

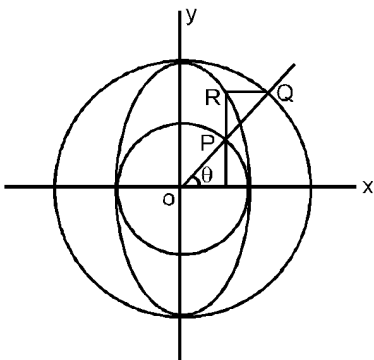
So ellipse touches both inner circle and outer circle

if foci are $(0, \pm a)$

$\Rightarrow a = be$ i.e $e = a/b$

$$\text{Also } e = \sqrt{1 - e^2} \Rightarrow e^2 = 1 - e^2 \Rightarrow e = \frac{1}{\sqrt{2}}$$

$$\text{and ratio of radii is } \frac{a}{b} = e = \frac{1}{\sqrt{2}}$$



8. Ans (3)

$$\lim_{x \rightarrow 0} ([f(x)] + x^2)^{\frac{1}{f(x)}}$$

$$= \lim_{x \rightarrow 0} (1 + x^2)^{\frac{1}{\frac{\tan x}{x} - 1}}$$

$$= \lim_{x \rightarrow 0} (1 + x^2)^{\frac{x}{\tan x - x}}$$

$$= \lim_{x \rightarrow 0} \frac{x^3}{\tan x - x} = \lim_{x \rightarrow 0} \frac{3x^2}{\sec^2 x - 1}$$

$$= \lim_{x \rightarrow 0} \frac{6x}{2\sec^2 x \tan x} = e^3$$

9. Ans (180)

$$f(x) = \begin{cases} ax^3 + b & , 0 \leq x \leq 1 \\ 2 \cos \pi x + \tan^{-1} x & , 1 < x \leq 2 \end{cases}$$

as $\tan [x^2]\pi = \tan n\pi$, $n \in \mathbb{I}$

$$f'(x) = \begin{cases} 3ax^2 & , 0 < x < 1 \\ -2\pi \sin \pi x + \frac{1}{1+x^2} & , 1 < x < 2 \end{cases}$$

As the function is differentiable in $[0, 2]$

$\Rightarrow$ function is differentiable at $x = 1$

$$\therefore f'(1^-) = f'(1^+)$$

$$\Rightarrow 3a = \frac{1}{2} \Rightarrow a = \frac{1}{6}$$

Function will also be continuous at $x = 1$

$$\therefore \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$$\Rightarrow a + b = -2 + \frac{\pi}{4}$$

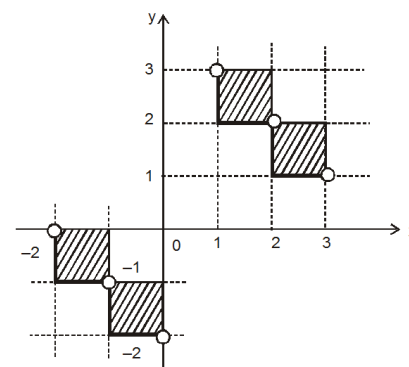
$$\therefore b = -2 - \frac{1}{6} + \frac{\pi}{4} = \frac{\pi}{4} - \frac{13}{6}$$

$$\Rightarrow k_1 = 6 \quad \& \quad k_2 = 12$$

$$\Rightarrow k_1^2 + k_2^2 = 180$$

10. Ans (4)

$$[x] \cdot [y] = 2$$



Here four cases arise

$$(1) [x] = 2 \& [y] = 1 \Rightarrow 2 \leq x < 3 \& 1 \leq y < 2$$

$$(2) [x] = 1 \& [y] = 2 \Rightarrow 1 \leq x < 2 \& 2 \leq y < 3$$

$$(3) [x] = -2 \& [y] = -1 \Rightarrow -2 \leq x < -1 \& -1 \leq y < 0$$

$$(4) [x] = -1 \& [y] = -2 \Rightarrow -1 \leq x < 0 \& -2 \leq y < -1$$

Area enclosed by solution set = 4



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

09-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	C	A	B	D	C	C	A	B
SECTION-II (i)	Q.	1	2	3	4	5	6		
	A.	1.50	2.00	2.50	6.02 to 6.03	200.00	20.00		
SECTION-II (ii)	Q.	7	8	9	10				
	A.	132	321	113	321				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	B	C	A	C	C	D	D	A
SECTION-II (i)	Q.	1	2	3	4	5	6		
	A.	4.00	4.00	2.00	6.91 to 6.93	7.75	3.00		
SECTION-II (ii)	Q.	7	8	9	10				
	A.	213	232	213	321				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	C	D	B	C	B	D	C	A
SECTION-II (i)	Q.	1	2	3	4	5	6		
	A.	3.00	2022.00	2000.00	12.00	0.00	1.00		
SECTION-II (ii)	Q.	7	8	9	10				
	A.	322	321	321	113				

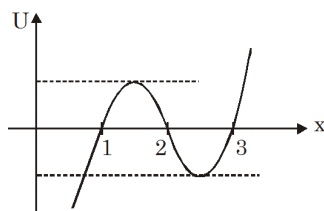
HINT – SHEET

PART-1 : PHYSICS

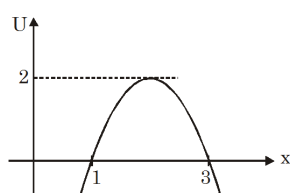
SECTION-I

1. Ans (C)

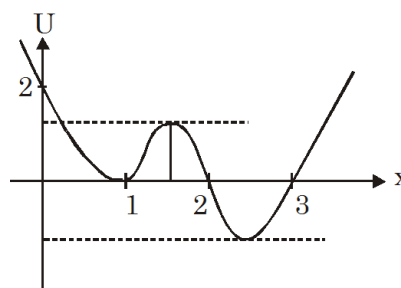
$$(I) \Rightarrow U(x) = \frac{1}{100} (x-1)(x-2)(x-3)$$



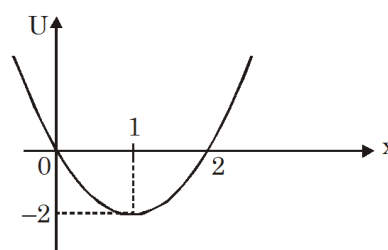
$$(II) U(x) = -2(x-1)(x-3)$$



$$(III) U(x) = \frac{1}{3} (x-1)^2 (x-2)(x-3)$$



$$(IV) U(x) = 2x(x-2)$$



$$F_x = -\frac{\partial U}{\partial x} > 0$$

$$\Rightarrow \frac{\partial U}{\partial x} < 0$$

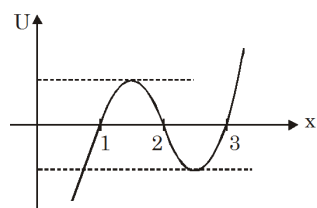
Stable equilibrium

$$\Rightarrow F = 0 \Rightarrow \frac{\partial U}{\partial x} = 0 \text{ and } \Rightarrow \frac{\partial^2 U}{\partial x^2} > 0$$

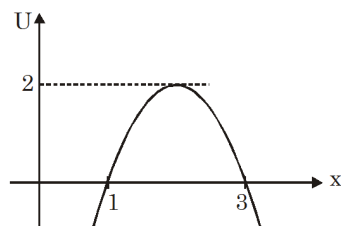
Now, this can be easily seen from graph of U vs x.

2. **Ans (A)**

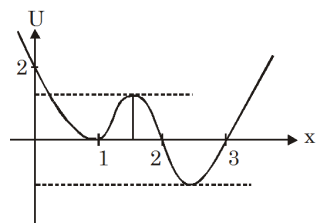
$$(I) \Rightarrow U(x) = \frac{1}{100} (x-1)(x-2)(x-3)$$



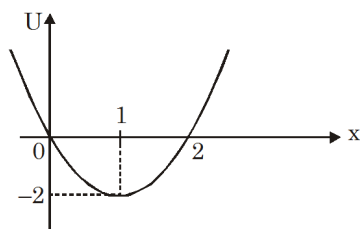
$$(II) U(x) = -2(x-1)(x-3)$$



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$$(IV) U(x) = 2x(x-2)$$



$$F_x = -\frac{\partial U}{\partial x} > 0$$

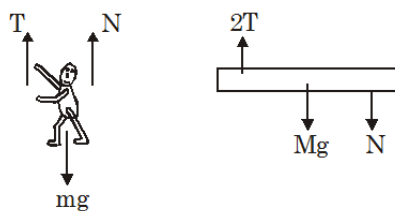
$$\Rightarrow \frac{\partial U}{\partial x} < 0$$

Stable equilibrium

$$\Rightarrow F = 0 \Rightarrow \frac{\partial U}{\partial x} = 0 \text{ and } \Rightarrow \frac{\partial^2 U}{\partial x^2} > 0$$

Now, this can be easily seen from graph of U vs x.

3. **Ans (B)**



$$3T = Mg + N$$

$$T + N = mg$$

$$3T = Mg + mg - T$$

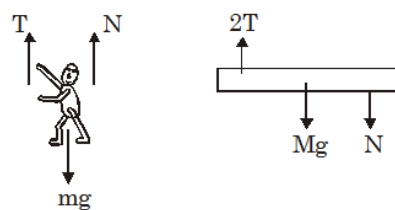
$$T = (M + m)g/4$$

$$N = (3m - M)g/4$$

$$w_2 = (3m - M)g/4$$

$$w_1 = 4T = (m + M)g$$

4. **Ans (D)**



$$T + N = mg$$

$$Mg + N = 2T$$

$$Mg + mg = 3T \Rightarrow T = (m + M) \frac{g}{3}$$

$$N = \frac{(2m - M)g}{3}$$

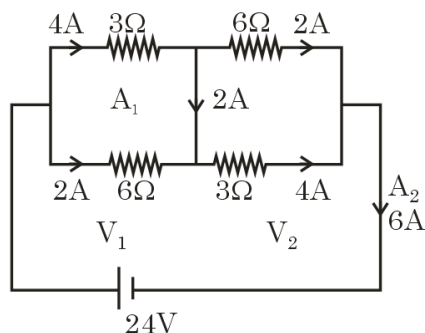
$$\text{by replacing } g \rightarrow g + \frac{g}{2}$$

$$N = w_2 = \frac{(2m - M)g}{2}$$

$$w_1 = 2T = (m + M) \times \frac{2g}{3} \times \frac{3}{2}$$

$$w_1 = (m + M)g$$

5. Ans (C)



$$R_{eq} = \frac{6 \times 3}{6 + 3} + \frac{6 \times 3}{6 + 3}$$

$$\Rightarrow R_{eq} = 4\Omega$$

$$i = \frac{E}{R_{eq}} = \frac{24V}{4\Omega} = 6A$$

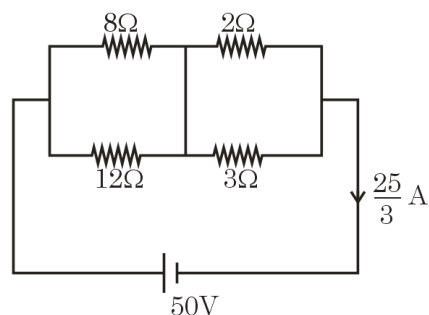
$$\therefore A_2 = 6A \text{ and } A_1 = 2A$$

$$\Rightarrow |A_2 - A_1| = 4A$$

Voltage across each 6Ω , $V_1 = 12V$ and voltage across each 3Ω , $V_2 = 12V$

$$\therefore |V_2 - V_1| = 0$$

6. Ans (C)



$$R_{eq} = \frac{12 \times 8}{12 + 8} + \frac{3 \times 2}{3 + 2} = 6\Omega$$

$$i = \frac{E}{R_{eq}} = \frac{50}{6} = \frac{25}{3}A$$

$$\therefore A_2 = \frac{25}{3}A$$

$$\text{Now, } V_1 = \frac{96}{20 \times 6} \times 50 = 40V \text{ and } V_2 = 50 - 40 = 10V$$

$$\therefore |V_1 - V_2| = 30$$

$$\text{and } A_1 = 0$$

$$\therefore |A_2 - A_1| = \frac{25}{3}$$

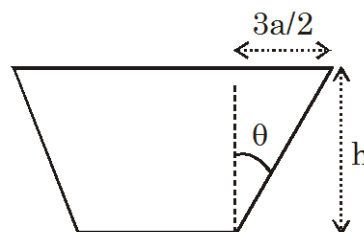
8. Ans (B)

When $\vec{a}$, $\vec{v}$ have same sign (direction) particle speed up, when they have opposite sign particle speed down.

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (1.50)



$$\tan \theta = \frac{3a}{2h}$$

$$2 \times 4a T \times \cos \theta = 4a \lambda g$$

$$\cos \theta = \frac{\lambda g}{2T}$$

$$\Rightarrow h = \frac{3\lambda ga}{2\sqrt{4T^2 - \lambda^2 g^2}}$$

2. Ans (2.00)

$$Ma = F_{net} = Mg - B - 6\pi\eta r v$$

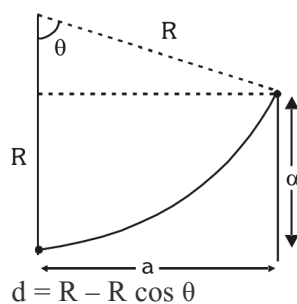
$$\text{at } v = v_T, a = 0$$

$$\Rightarrow Mg - B = 6\pi\eta r v_T$$

$$\Rightarrow v_T = \frac{\frac{4}{3}\pi r^3 \left[\rho - \frac{\rho}{2}\right] g}{6\pi\eta} = 2m/\text{sec}$$

3. Ans (2.50)

$$a = R \sin \theta$$



$$\therefore \left(\frac{a}{R}\right)^2 + \left(\frac{d-R}{R}\right)^2 = 1$$

$$\Rightarrow a^2 + d^2 + R^2 - 2Rd = R^2$$

$$\Rightarrow \frac{a^2 + d^2}{2d} = R = \frac{mv}{qB}$$

$$mv = qB \left(\frac{a^2 + d^2}{2d}\right)$$

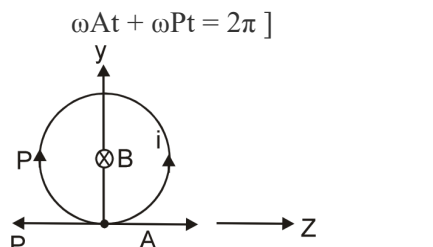
$$mv = 0.32 \left(\frac{3^2 + 4^2}{2 \times 4}\right)$$

$$mv = \frac{0.32}{8} (25)$$

$$mv = 1$$

4. Ans (6.02 to 6.03)

A and P will have the same momentum in magnitude and they will move in opposite directions. They will move in the circle of same radius and the same centre but in opposite directions. If they meet after time t then

$$\omega_A t + \omega_P t = 2\pi$$


$$\Rightarrow t = \frac{2\pi}{\omega_A + \omega_P} = \frac{2\pi}{\frac{2eB}{4m} + \frac{2eB}{(A-4)m}}$$

$$t = \frac{4(A-4)m\pi}{eBA}; \theta_A = \omega_A t$$

$$= \frac{2eB}{4m} \times \frac{4m(A-4)}{eBA}$$

$$= \frac{2(A-4)\pi}{A} = \frac{48}{25}\pi \Rightarrow n = 48$$

5. Ans (200.00)

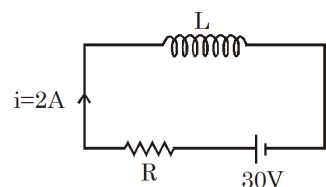
Let they meet after time t sec.

$$V_1 \cos 45^\circ t = V_2 \cos 60^\circ t = a$$

$$\text{So } V_1 = \frac{a}{\cos 45^\circ}, V_2 = \frac{a}{t \cos 60^\circ}$$

$$\text{for vertical } (V_2 \sin 60^\circ - V_1 \sin 45^\circ) t = h$$

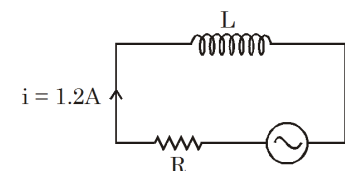
6. Ans (20.00)



$$i = \frac{V}{R}$$

$$2 = \frac{30}{R}$$

$$R = 15\Omega$$



$$V_{rms} = I_{rms} Z$$

$$30 = 1.2 Z$$

$$Z = 25\Omega$$

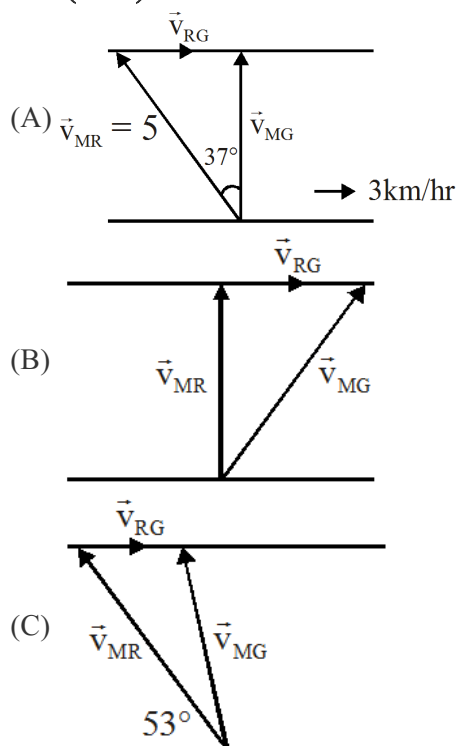
$$\sqrt{(\omega L)^2 + R^2} = 25$$

$$X_L = \omega L = 20\Omega$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (132)



10. Ans (321)

$$\text{For (A) } d = \pi - 2i$$

$$\text{For (B) : } \delta = r - i = \sin^{-1} \left[\frac{\mu_1 \sin i}{\mu_2} \right] - i$$

PART-2 : CHEMISTRY

SECTION-II (i)

2. Ans (4.00)

Carbyl amine reaction gives foul smell with 1° amine and aniline.

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$(1 - x^2) \frac{dy}{dx} + xy = ax$$

$$\frac{dy}{dx} + \left(\frac{x}{1 - x^2} \right) y = \frac{ax}{1 - x^2}$$

$$\text{I.F.} = e^{\int \frac{x}{1-x^2} dx} = e^{-\frac{1}{2} \ln(1-x^2)} = \begin{cases} \frac{1}{\sqrt{1-x^2}}, & -1 < x < 1 \\ \frac{1}{\sqrt{x^2-1}}, & x^2 > 1 \end{cases}$$

$$(I) x \in (-1, 1)$$

$$\frac{y}{\sqrt{1-x^2}} = \int \frac{ax dx}{\sqrt{1-x^2} (1-x^2)}$$

$$= a \int \frac{x}{(1-x^2)^{3/2}} dx = \left(-\frac{a}{2} \right) \left(-\frac{2}{\sqrt{1-x^2}} \right) + C$$

$$\frac{y}{\sqrt{1-x^2}} = \frac{a}{\sqrt{1-x^2}} + C$$

$$y = a + C\sqrt{1-x^2}$$

$$C^2(1-x^2) = (y-a)^2$$

$$C^2x^2 + (y-a)^2 = C^2 \quad \text{ellipse}$$

$$(II) x^2 > 1 \Rightarrow \text{similarly solving}$$

$$C^2x^2 - (y-a)^2 = C^2$$

2. Ans (D)

$$(1 - x^2) \frac{dy}{dx} + xy = ax$$

$$\frac{dy}{dx} + \left(\frac{x}{1 - x^2} \right) y = \frac{ax}{1 - x^2}$$

$$\text{I.F.} = e^{\int \frac{x}{1-x^2} dx} = e^{-\frac{1}{2} \ln(1-x^2)} = \begin{cases} \frac{1}{\sqrt{1-x^2}}, & -1 < x < 1 \\ \frac{1}{\sqrt{x^2-1}}, & x^2 > 1 \end{cases}$$

$$(I) x \in (-1, 1)$$

$$\frac{y}{\sqrt{1-x^2}} = \int \frac{ax dx}{\sqrt{1-x^2} (1-x^2)}$$

$$= a \int \frac{x}{(1-x^2)^{3/2}} dx = \left(-\frac{a}{2} \right) \left(-\frac{2}{\sqrt{1-x^2}} \right) + C$$

$$\frac{y}{\sqrt{1-x^2}} = \frac{a}{\sqrt{1-x^2}} + C$$

$$y = a + C\sqrt{1-x^2}$$

$$C^2(1-x^2) = (y-a)^2$$

$$C^2x^2 + (y-a)^2 = C^2 \quad \text{ellipse}$$

$$(II) x^2 > 1 \Rightarrow \text{similarly solving}$$

$$C^2x^2 - (y-a)^2 = C^2$$

3. Ans (B)

Dices are alike

number of ways for sum to be 6

(1, 1, 4), (1, 2, 3), (2, 2, 2)

number of ways = 3

for calculating probability, we will consider the dice as distinct

$$\text{so probability} = \frac{3 + 6 + 1}{216} = \frac{10}{216} = \frac{5}{108}$$

number of ways for sum to be 15 will be same as number of ways for sum to be 6

$$\text{probability} = \frac{5}{108}$$

total number of outcome when dices are alike

$$= {}^6C_3 + {}^6C_2 \times 2 + 6 = 20 + 30 + 6 = 56$$

number of ways in which a sum less than 11 appear = 28

same as number of ways in which sum ≥ 11 appear

$$\text{probability} = \frac{1}{2}$$

4. Ans (C)

Dices are alike

number of ways for sum to be 6

(1, 1, 4), (1, 2, 3), (2, 2, 2)

number of ways = 3

for calculating probability, we will consider the dice as distinct

$$\text{so probability} = \frac{3 + 6 + 1}{216} = \frac{10}{216} = \frac{5}{108}$$

number of ways for sum to be 15 will be same as number of ways for sum to be 6

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total number of outcome when dices are alike

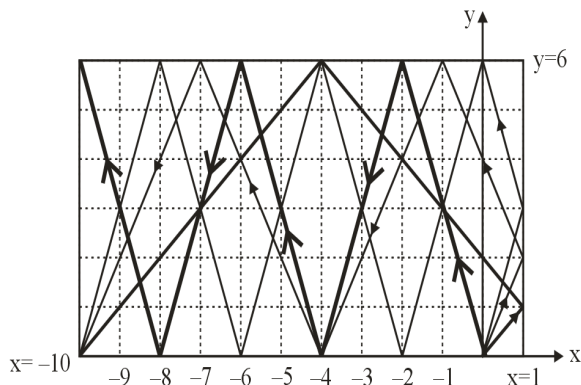
$$= {}^6C_3 + {}^6C_2 \times 2 + 6 = 20 + 30 + 6 = 56$$

number of ways in which a sum less than 11 appear = 28

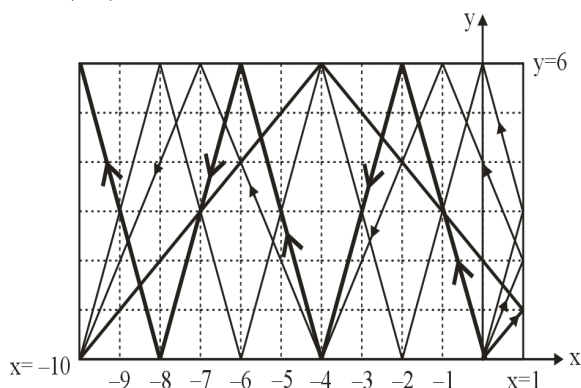
same as number of ways in which sum ≥ 11 appear

$$\text{probability} = \frac{1}{2}$$

5. Ans (B)



6. Ans (D)



7. Ans (C)

Center of such circle (h, h) , radius $= h$

equation : $x^2 + y^2 - 2hx - 2hy + h^2 = 0$

passes through (a, b)

$$a^2 + b^2 - 2ah - 2bh + h^2 = 0$$

$$h^2 - 2(a+b)h + a^2 + b^2 = 0 \begin{cases} h_1 \\ h_2 \end{cases}$$

Two such circles

$$h_1 + h_2 = 2(a+b), h_1 h_2 = a^2 + b^2$$

centers $(h_1, h_1), (h_2, h_2)$, sum of ordinates $= h_1 + h_2$

$$= 2(a+b)$$

$$\text{dist}^n \text{ between centers} = \sqrt{2} |h_1 - h_2|$$

$$= \sqrt{2} \sqrt{4(a+b)^2 - 4(a^2 + b^2)}$$

$$= 2\sqrt{2} \sqrt{2ab} = 4\sqrt{ab}$$

8. Ans (A)

Center of such circle (h, h) , radius $= h$

$$\text{equation : } x^2 + y^2 - 2hx - 2hy + h^2 = 0$$

passes through (a, b)

$$a^2 + b^2 - 2ah - 2bh + h^2 = 0$$

$$h^2 - 2(a+b)h + a^2 + b^2 = 0 \begin{cases} h_1 \\ h_2 \end{cases}$$

Two such circles

$$h_1 + h_2 = 2(a+b), h_1 h_2 = a^2 + b^2$$

centers $(h_1, h_1), (h_2, h_2)$, sum of ordinates

$$= h_1 + h_2 = 2(a+b)$$

$$\text{dist}^n \text{ between centers} = \sqrt{2} |h_1 - h_2|$$

$$= \sqrt{2} \sqrt{4(a+b)^2 - 4(a^2 + b^2)}$$

$$= 2\sqrt{2} \sqrt{2ab} = 4\sqrt{ab}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (3.00)

$$f(n+1) - f(n) = \int_0^{\pi/2} \frac{\sin^2(n+1)\theta - \sin^2 n\theta}{\sin^2 \theta} d\theta$$

$$= \int_0^{\pi/2} \frac{\sin(2n+1)\theta}{\sin \theta} d\theta = g(n)$$

$$g(n+1) - g(n) = \int_0^{\pi/2} \frac{\sin[(2n+3)\theta] - \sin[(2n+1)\theta]}{\sin \theta} d\theta$$

$$= \int_0^{\pi/2} \frac{2 \cos((2n+2)\theta) \sin \theta}{\sin \theta} d\theta$$

$$= 0$$

$$g(n+1) = g(n) = g(n-1) = \dots = \frac{\pi}{2}$$

$$f(1) = \frac{\pi}{2}$$

$$g(2) = \pi, f(n) = \frac{n\pi}{2}$$

$$\frac{f(15) + f(3)}{f(15) - f(9)} = \frac{\frac{15}{2} + \frac{3}{2}}{\frac{15}{2} - \frac{9}{2}} = 3$$

2. Ans (2022.00)

$$\lim_{n \rightarrow \infty} n^2 \int_0^{\frac{1}{n}} (4044) \cos x \cdot x \, dx$$

$$\lim_{n \rightarrow \infty} \frac{(4044) \cos\left(\frac{1}{n}\right) \cdot \left(\frac{1}{n}\right) \left(-\frac{1}{n^2}\right)}{-\left(\frac{2}{n^3}\right)}$$

2022

3. Ans (2000.00)

$$\text{Let } \vec{p} = 3\hat{i} + 4\hat{j} + 10\hat{k}$$

$$\vec{q} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\vec{p} \cdot \vec{q} = |\vec{p}| |\vec{q}| \cos \theta$$

$$3a + 4b + 10c = \sqrt{125} \sqrt{(a^2 + b^2 + c^2)} \cos \theta$$

$$\sqrt{N} = \sqrt{125} \sqrt{16} \cdot 1$$

$$N = 125 \cdot 16 = 2000$$

4. Ans (12.00)

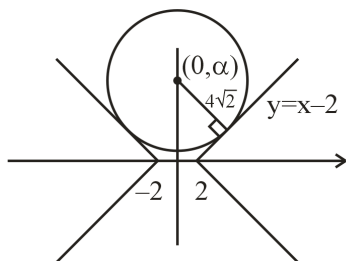
$$\frac{|\alpha + 2|}{\sqrt{2}} = 4\sqrt{2}$$

$$|\alpha + 2| = 8$$

$$\alpha \neq -10, \alpha = 6$$

$$\text{Similary } \alpha = -6$$

$$\text{Sum} = 12$$



5. Ans (0.00)

$$-2f(x) \leq f'(x) \leq 2f(x)$$

$$f'(x) + 2f(x) \geq 0, f'(x) - 2f(x) \leq 0$$

$$\frac{d}{dx} (e^{2x} f(x)) \geq 0, \frac{d}{dx} (e^{-2x} f(x)) \leq 0$$

$$e^{2x} f(x) \geq 0, e^{-2x} f(x) \leq 0$$

$$f(x) = 0 \forall x \in [1, 5]$$

6. Ans (1.00)

$$\text{Let } |C| \neq 0$$

$$AC - CA = C^2$$

$$C^{-1} AC C^{-1} - C^{-1} CA C^{-1} = I$$

$$C^{-1} A - AC^{-1} = I$$

which is not possible as

$$\text{tr}(C^{-1} A - AC^{-1}) = 0$$

$$\text{hence } |C| = 0$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (322)

$$f(x) = 7 + 10x \ln 5 - \frac{1}{5} \cdot 5^x - 25 \cdot 5^{-x}$$

$$f'(x) = 10 \ln 5 - \frac{5^x}{5} \ln 5 + \frac{25}{5^x} \ln 5$$

$$= -\frac{\ln 5}{5 \cdot 5^x} [5^{2x} - 25] (5^x + 5)$$

maximum at $x = 2$

$$r = \prod_{x \rightarrow 0}^{a=2} \frac{\int_0^x t^2 \, dt}{x^2 \tan x} = \frac{1}{3}$$

$$S = \frac{2}{1 - \frac{1}{3}} = 3$$

8. Ans (321)

Let sides QR, RP, PQ are $n, n+1, n+2$

(P) $\angle R = 2\angle P$

$$\frac{\sin 2P}{n+2} = \frac{\sin P}{n}$$

$$2 \cos P = \left(\frac{n+2}{n} \right)$$

$$2 \left[\frac{(n+2)^2 + (n+1)^2 - n^2}{2(n+2)(n+1)} \right] = \frac{n+2}{n}$$

$$[n^2 + 6n + 5]n = (n+2)^2(n+1)$$

$$n^3 + 6n^2 + 5n = n^3 + 5n^2 + 8n + 4$$

$$n^2 - 3n - 4 = 0 \Rightarrow (n-4)(n+1) = 0 \Rightarrow n = 4$$

biggest side = 6

(Q) $\vec{PR} \cdot \vec{RQ} = 0$

$$(n+2)^2 = n^2 + (n+1)^2$$

$$n^2 + 4n + 4 = 2n^2 + 2n + 1$$

$$n^2 - 2n - 3 = 0 \Rightarrow n = 3$$

$$\text{Area of } \Delta PQR = \frac{1}{2}$$

$$|\vec{p} \times \vec{q} + \vec{q} \times \vec{r} + \vec{r} \times \vec{p}| = \frac{1}{2} \times 3 \times 4$$

$$(R) \left| \frac{\left(-\frac{a_1}{b_1}\right) - \left(-\frac{a_2}{b_2}\right)}{1 + \frac{a_1 a_2}{b_1 b_2}} \right| = \tan P = \frac{4}{3}$$

$$\cos P = \frac{3}{5} = \frac{(n+2)^2 + (n+1)^2 - n^2}{2(n+1)(n+2)}$$

$$n = 13$$

$$\text{semiperimeter} = \frac{13 + 14 + 15}{2} = 21$$

9. Ans (321)

$$g(x) = Ax^2 + Bx + C, g(0) = -6, g'(0) = C, g''(0) = -2b$$

$$g'(x) = 2Ax + B, g''(x) = 2A$$

$$g(x) = \frac{1}{2} g''(0) x^2 + g'(0) \cdot x - 6 = -bx^2 + cx - 6$$

$$h(x) = g(x) \Rightarrow 4x^4 - ax^3 = -bx^2 + cx - 6$$

$$4x^4 - ax^3 + bx^2 - cx + 6 = 0 \rightarrow x_1, x_2, x_3, x_4$$

$$\frac{\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4}}{4} \geq \left(\frac{24}{x_1 x_2 x_3 x_4} \right)^{1/4}$$

$$2 \geq \left(\frac{24}{3/2} \right)^{1/4} \Rightarrow 2 \geq 2$$

$$\frac{1}{x_1} = \frac{2}{x_2} = \frac{3}{x_3} = \frac{4}{x_4} = \frac{1}{k} \Rightarrow x_1 = k, x_2 = 2k,$$

$$x_3 = 3k, x_4 = 4k$$

$$\frac{1}{k} + \frac{1}{k} + \frac{1}{k} + \frac{1}{k} = 8 \Rightarrow k = \frac{1}{2}$$

$$x_1 = \frac{1}{2}, x_2 = 1, x_3 = \frac{3}{2}, x_4 = 2$$

$$\frac{a}{4} = 5 \Rightarrow a = 20, \frac{b}{4} = \frac{1}{2} + \frac{3}{4} + 1 + \frac{3}{2} + 2 + 3$$

$$\boxed{b = 35}$$

$$\frac{c}{4} = \frac{3}{4} + 3 + 1 + \frac{3}{2} \in \boxed{c = 25}$$

10. Ans (113)

$$(A) f(0) = 1, f(0^-) = 1, f(0^+) = 1$$

contⁿ at $x = 0$

discontⁿ at $x = -1, -2$

$$f(1) = \frac{1}{2}, f(1^+) = 0, f(1^-) = 1$$

discontⁿ at $x = 1$

for $x > 1$ $f(x) = 0$ constant

contⁿ at $x = 0, 2, 3, 4, 5, 6, 7, 8, 9$

$$(B) \int_1^2 (2-x) f''(x) dx = 7, f(1) = -1, f'(1) = 3$$

$$(f(x)(2-x)) \Big|_1^2 - \int_1^2 f'(x)(-x) dx = 7$$

$$0 - f'(1) + \int_1^2 f'(x) dx = 7$$

$$f(2) - f(1) = 10 \Rightarrow f(2) = 9$$

$$(C) \int_a^b (4+3x-x^2) dx = \int_a^b -(x-4)(x+1) dx$$

$$a = -1, b = 4$$

$$a + b = 3$$

